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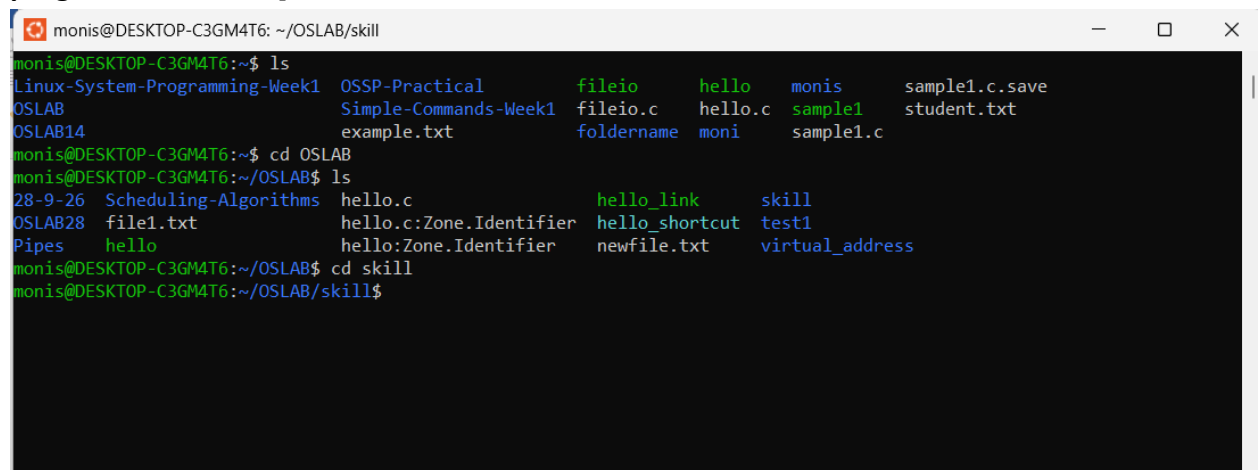
Section:6

ForgeOS/Shellforge/

OWN SHELL

1. Create your own shell

[Open Ubuntu and go to your Skill directory. Use the directory where your 24 Skill programs are saved:]



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSSP-Practical      fileio      hello      monis      sample1.c.save
OSLAB                          Simple-Commands-Week1 fileio.c    hello.c    sample1    student.txt
OSLAB14                        example.txt         foldername  moni       sample1.c
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
28-9-26  Scheduling-Algorithms  hello.c      hello_link  skill
OSLAB28  file1.txt              hello.c:Zone.Identifier  hello_shortcut  test1
Pipes    hello                  hello:Zone.Identifier    newfile.txt    virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSSP-Practical      fileio      hello      monis      sample1.c.save
OSLAB                          Simple-Commands-Week1 fileio.c     hello.c     sample1     student.txt
OSLAB14                        example.txt          foldername  moni       sample1.c
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
28-9-26  Scheduling-Algorithms  hello.c      hello_link  skill
OSLAB28  file1.txt              hello.c:Zone.Identifier  hello_shortcut  test1
Pipes    hello                  hello:Zone.Identifier    newfile.txt    virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls
README_Q24.md      double_quotes      list_jobs.c      pass_arguments.c  signal_handlers
actual_output.txt  double_quotes.c    long_pipeline     path_missing       signal_handlers.c
append.txt         error.txt          long_pipeline.c   path_missing.c     sigtstp_jobs
append_redirection error_handling      memory            path_resolution    sigtstp_jobs.c
append_redirection.c error_handling.c    memory.4         path_resolution.c  sigtstp_jobs.c
architecture.txt   escape_parser      memory.c          performance.txt     single_pipe
background_jobs    escape_parser.c    memory1           performance_output.txt single_pipe.c
background_jobs.c  execution_error    memory1.c        pipe_sync           single_quotes
build_Q24.sh       execution_error.c  memory2           pipe_sync.c        single_quotes.c
builtin_dispatch   expected_output.txt memory2.c         pipeline_output.txt source_files.txt
builtin_dispatch.c export_child        memory3           pipeline_structure  stderr_redirection
builtins           export_child.c     memory3.c        pipeline_structure.c stderr_redirection.c
builtins.c         export_variables   memory3.c.save   process             stress_report.txt
child_process      export_variables.c memory4           process.c           stress_test.sh
child_process.c    failure_output.txt memory4.c        process_groups      test.sh
combined.txt       final_q15.txt      memory_test       process_groups.c   test_report.txt
combined_redirection foreground_switch   memory_test.c     process_groups.c.save tokenization
combined_redirection.c foreground_switch.c multiple_pipes     process_pipeline   tokenization.c
command_history     history            multiple_pipes.c  process_pipeline.c tokenization.c.save
command_history.c   history.c          no_match.txt      process_pipeline.c tokenization.c
command_launch      input.txt          output.txt        q21_errors.log     valgrind_report.txt
command_launch.c    input1.txt        output1.txt       resume_jobs         variable_expansion
complex_commands    input_redirection output_redirection shell               variable_expansion.c
complex_commands.c  input_redirection.c output_redirection.c shell.c            waitpid_monitor
complex_commands1.c integrated_shell   parent_child      sigint_forwarding  waitpid_monitor.c
directory_navigation integrated_shell.c  parent_child.c    sigint_forwarding.c
directory_navigation.c job_table          parser_logic       sigint_forwarding.c.save
directory_state     job_table.c        parser_logic.c     signal_behavior     signal_behavior.c
directory_state.c   list_jobs          pass_arguments     signal_behavior.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$

```

cd ~/OSLAB/skill

nano myshell.c

```

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano myshell.c

```

Paste this code:

```

#define _POSIX_C_SOURCE 200809L

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

#define MAX_ARGS 64

```

```

int main(void)
{
    char *line = NULL;
    size_t capacity = 0;

    printf("=====\n");
    printf(" Welcome to 2520030537 SHELLFORGE\n");
    printf("=====\n");

    while (1) {
        printf("2520030537_SHELLFORGE$ ");
        fflush(stdout);

        if (getline(&line, &capacity, stdin) == -1) {
            printf("\n");
            break;
        }

        char *args[MAX_ARGS];
        int count = 0;

        char *token = strtok(line, " \t\n");

        while (token != NULL && count < MAX_ARGS - 1) {
            args[count++] = token;
            token = strtok(NULL, " \t\n");
        }

        args[count] = NULL;

        if (count == 0)
            continue;

        if (strcmp(args[0], "exit") == 0) {
            printf("Exiting SHELLFORGE...\n");
            break;
        }

        if (strcmp(args[0], "help") == 0) {
            printf("Supported commands:\n");
            printf("  ls, pwd, date, cal, ps\n");
            printf("  gcc, cat, mkdir and other Linux commands\n");
        }
    }
}

```

```

        printf("  cd <directory> - change directory\n");
        printf("  ./program      - run a compiled program\n");
        printf("  exit          - leave the shell\n");
        continue;
    }

    if (strcmp(args[0], "cd") == 0) {
        const char *path =
            count > 1 ? args[1] : getenv("HOME");

        if (path == NULL || chdir(path) != 0)
            perror("cd");

        continue;
    }

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        continue;
    }

    if (pid == 0) {
        execvp(args[0], args);
        perror("Command failed");
        _exit(127);
    }

    int status;

    if (waitpid(pid, &status, 0) == -1)
        perror("waitpid");
}

free(line);
return 0;
}

```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 myshell.c *
#define _POSIX_C_SOURCE 200809L

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

#define MAX_ARGS 64

int main(void)
{
    char *line = NULL;
    size_t capacity = 0;

    printf("=====\n");
    printf(" Welcome to 2520030537 SHELLFORGE\n");
    printf("=====\n");

    while (1) {
        printf("2520030537_SHELLFORGE$ ");
        fflush(stdout);

        if (getline(&line, &capacity, stdin) == -1) {
            printf("\n");
            break;
        }

        char *args[MAX_ARGS];
        int count = 0;

        char *token = strtok(line, " \t\n");

        while (token != NULL && count < MAX_ARGS - 1) {
            args[count++] = token;
            token = strtok(NULL, " \t\n");
        }

        args[count] = NULL;

        if (count == 0)
            continue;

        if (strcmp(args[0], "exit") == 0) {
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 myshell.c *
if (strcmp(args[0], "exit") == 0) {
    printf("Exiting SHELLFORGE...\n");
    break;
}

if (strcmp(args[0], "help") == 0) {
    printf("Supported commands:\n");
    printf("  ls, pwd, date, cal, ps\n");
    printf("  gcc, cat, mkdir and other Linux commands\n");
    printf("  cd <directory> - change directory\n");
    printf("  ./program      - run a compiled program\n");
    printf("  exit           - leave the shell\n");
    continue;
}

if (strcmp(args[0], "cd") == 0) {
    const char *path =
        count > 1 ? args[1] : getenv("HOME");

    if (path == NULL || chdir(path) != 0)
        perror("cd");

    continue;
}

pid_t pid = fork();

if (pid < 0) {
    perror("fork");
    continue;
}

if (pid == 0) {
    execvp(args[0], args);
    perror("Command failed");
    _exit(127);
}

int status;

if (waitpid(pid, &status, 0) == -1)
    perror("waitpid");

free(line);

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 myshell.c *
}

if (strcmp(args[0], "help") == 0) {
    printf("Supported commands:\n");
    printf("  ls, pwd, date, cal, ps\n");
    printf("  gcc, cat, mkdir and other Linux commands\n");
    printf("  cd <directory> - change directory\n");
    printf("  ./program      - run a compiled program\n");
    printf("  exit           - leave the shell\n");
    continue;
}

if (strcmp(args[0], "cd") == 0) {
    const char *path =
        count > 1 ? args[1] : getenv("HOME");

    if (path == NULL || chdir(path) != 0)
        perror("cd");

    continue;
}

pid_t pid = fork();

if (pid < 0) {
    perror("fork");
    continue;
}

if (pid == 0) {
    execvp(args[0], args);
    perror("Command failed");
    _exit(127);
}

int status;

if (waitpid(pid, &status, 0) == -1)
    perror("waitpid");
}

free(line);
return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo
```

Write to File: myshell.c

^G Help M-D DOS Format M-A Append M-B Backup File ^T Browse
^C Cancel M-M Mac Format M-P Prepend ^O Discard buffer

2. Compile and run your shell

Run these commands in the normal Ubuntu terminal:

```
gcc -Wall -Wextra myshell.c -o myshell
```

```
./myshell
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano myshell.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -Wall -Wextra myshell.c -o myshell
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./myshell
=====
Welcome to 2520030537 SHELLFORGE
=====
2520030537_SHELLFORGE$
```

Now enter commands at your custom prompt:

```
2520030537_SHELLFORGE$ pwd
2520030537_SHELLFORGE$ ls
2520030537_SHELLFORGE$ date
2520030537_SHELLFORGE$ help
```

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill

monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$./myshell

```
=====
Welcome to 2520030537 SHELLFORGE
=====
2520030537_SHELLFORGE$ pwd
/home/monis/OSLAB/skill
2520030537_SHELLFORGE$ ls
README_Q24.md          execution_error        memory4.c             q21_errors.log
actual_output.txt      execution_error.c     memory_test           resume_jobs
append.txt             expected_output.txt  memory_test.c        resume_jobs.c
append_redirection     export_child          multiple_pipes        shell
append_redirection.c   export_child.c        multiple_pipes.c      shell.c
architecture.txt       export_variables      myshell              sigint_forwarding
background_jobs         export_variables.c    myshell.c            sigint_forwarding.c
background_jobs.c       failure_output.txt    no_match.txt          sigint_forwarding.c.save
build_Q24.sh           final_q15.txt         output.txt            signal_behavior
builtin_dispatch        foreground_switch     output1.txt           signal_behavior.c
builtin_dispatch.c      foreground_switch.c   output_redirection    signal_handlers
builtins               history              output_redirection.c  signal_handlers.c
builtins.c             history.c             parent_child          sigtstp_jobs
child_process          input.txt             parent_child.c        sigtstp_jobs.c
child_process.c        input1.txt            parser_logic          single_pipe
combined.txt           input_redirection     pass_arguments        single_pipe.c
combined_redirection   input_redirection.c  pass_arguments.c      single_quotes
combined_redirection.c integrated_shell       path_missing          single_quotes.c
command_history        integrated_shell.c    path_missing.c        source_files.txt
command_history.c      job_table             path_resolution       stderr_redirection
command_launch         job_table.c           path_resolution.c     stderr_redirection.c
command_launch.c       list_jobs             performance.txt       stress_report.txt
complex_commands       list_jobs.c           performance_output.txt stress_test.sh
complex_commands.c     long_pipeline         pipe_sync              test.sh
complex_commands1.c    long_pipeline.c       pipe_sync.c           test_report.txt
directory_navigation   memory                pipeline_output.txt   tokenization
directory_navigation.c memory.4               pipeline_structure    tokenization.c
directory_state        memory.c              pipeline_structure.c  tokenization.c.save
directory_state.c      memory1               process               tokenization.c
double_quotes          memory1.c             process.c             valgrind_report.txt
double_quotes.c        memory2               process_groups        variable_expansion
error.txt              memory2.c             process_groups.c      variable_expansion.c
error_handling         memory3               process_groups.c.save waitpid_monitor
error_handling.c       memory3.c             process_pipeline      waitpid_monitor.c
escape_parser          memory3.c.save        process_pipeline.c
escape_parser.c        memory4               process_pipeline.c
```

```

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
complex_commands1.c    long_pipeline.c    pipe_sync          test.sh
directory_navigation  memory             pipe_sync.c        test_report.txt
directory_navigation.c memory.4            pipeline_output.txt tokenization
directory_state       memory.c           pipeline_structure  tokenization.c
directory_state.c     memory1            pipeline_structure.c tokenization.c.save
double_quotes         memory1.c          process             tokenization.c
double_quotes.c       memory2            process.c           valgrind_report.txt
error.txt             memory2.c          process_groups      variable_expansion
error_handling        memory3            process_groups.c    variable_expansion.c
error_handling.c      memory3.c          process_groups.c.save waitpid_monitor
escape_parser         memory3.c.save     process_pipeline    waitpid_monitor.c
escape_parser.c       memory4            process_pipeline.c

2520030537_SHELLFORGE$ date
Tue Sep 29 18:25:19 IST 2026
2520030537_SHELLFORGE$ help
Supported commands:
  ls, pwd, date, cal, ps
  gcc, cat, mkdir and other Linux commands
  cd <directory> - change directory
  ./program      - run a compiled program
  exit           - leave the shell
2520030537_SHELLFORGE$
2520030537_SHELLFORGE$

```

→cal is failing because the calendar command is probably not installed in your Ubuntu environment

Step 1 – Exit ShellForge

At your current prompt, type:

exit

```

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
2520030537_SHELLFORGE$ cal
Command failed: No such file or directory
2520030537_SHELLFORGE$ exit
Exiting SHELLFORGE...
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$

```

Step 2 – Install the calendar command

In your normal Ubuntu terminal, run:

sudo apt update

sudo apt install ncal

Enter your Ubuntu password if prompted.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ sudo apt update
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [543 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [679 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-updates/main Translation-en [162 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.0 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [308 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [98.9 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [190 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [127 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [187 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [59.3 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [443 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [87.8 kB]
Fetched 3485 kB in 3s (1137 kB/s)
51 packages can be upgraded. Run 'apt list --upgradable' to see them.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ sudo apt install ncal
Installing:
  ncal

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 51
  Download size: 20.8 kB
  Space needed: 59.4 kB / 1023 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 ncal amd64 12.1.8build1 [20.8 kB]
Fetched 20.8 kB in 1s (26.6 kB/s)
Selecting previously unselected package ncal.
(Reading database ... 43085 files and directories currently installed.)
Preparing to unpack .../ncal_12.1.8build1_amd64.deb ...
Unpacking ncal (12.1.8build1) ...
Setting up ncal (12.1.8build1) ...
Processing triggers for man-db (2.13.1-1build1) ...
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 3 – Check whether cal works

cal

You should see a calendar for the current month.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ cal
September 2026
Su Mo Tu We Th Fr Sa
    1  2  3  4  5
 6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30

monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 4 – Restart your ShellForge

Go to the folder containing your shell program and run:

`./myshell`

Enter:

2520030537_SHELLFORGE\$ `ls`

2520030537_SHELLFORGE\$ `pwd`

2520030537_SHELLFORGE\$ `date`

2520030537_SHELLFORGE\$ `cal`

2520030537_SHELLFORGE\$ `ps`

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill

monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$./myshell

```
=====
Welcome to 2520030537 SHELLFORGE
=====
2520030537_SHELLFORGE$ ls
README_Q24.md          execution_error      memory4.c           q21_errors.log
actual_output.txt      execution_error.c   memory_test         resume_jobs
append.txt             expected_output.txt memory_test.c       resume_jobs.c
append_redirection     export_child        multiple_pipes      shell
append_redirection.c   export_child.c      multiple_pipes.c    shell.c
architecture.txt       export_variables    myshell            sigint_forwarding
background_jobs         export_variables.c  myshell.c          sigint_forwarding.c
background_jobs.c       failure_output.txt  no_match.txt       sigint_forwarding.c.save
build_Q24.sh           final_q15.txt       output.txt          signal_behavior
builtin_dispatch        foreground_switch   output1.txt         signal_behavior.c
builtin_dispatch.c      foreground_switch.c output_redirection  signal_handlers
builtins               history            output_redirection.c signal_handlers.c
builtins.c             history.c          parent_child        sigtstp_jobs
child_process          input.txt          parent_child.c      sigtstp_jobs.c
child_process.c        input1.txt         parser_logic        single_pipe
combined.txt           input_redirection  parser_logic.c      single_pipe.c
combined_redirection   input_redirection.c pass_arguments      single_quotes
combined_redirection.c integrated_shell     path_missing        single_quotes.c
command_history        integrated_shell.c  path_missing.c      source_files.txt
command_history.c      job_table          path_resolution     stderr_redirection
command_launch         job_table.c        path_resolution.c   stderr_redirection.c
command_launch.c       list_jobs          performance.txt     stress_report.txt
complex_commands       list_jobs.c        performance_output.txt stress_test.sh
complex_commands.c     long_pipeline      pipe_sync           test.sh
complex_commands1.c    long_pipeline.c    pipe_sync.c         test_report.txt
directory_navigation   memory             pipeline_output.txt tokenization
directory_navigation.c memory.4            pipeline_structure  tokenization.c
directory_state        memory.c           pipeline_structure.c tokenization.c.save
directory_state.c      memory1            process            tokenization.c
double_quotes          memory1.c          process.c           valgrind_report.txt
double_quotes.c        memory2.c          process_groups      variable_expansion
error.txt              memory3            process_groups.c    variable_expansion.c
error_handling         memory3.c          process_groups.c.save waitpid_monitor
error_handling.c       memory3.c.save     process_pipeline    waitpid_monitor.c
escape_parser          memory4            process_pipeline.c
escape_parser.c
2520030537_SHELLFORGE$ pwd
/home/monis/OSLAB/skill
2520030537_SHELLFORGE$ date
Tue Sep 29 18:37:40 IST 2026
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
error_handling      memory3             process_groups.c    variable_expansion.c
error_handling.c    memory3.c           process_groups.c.save waitpid_monitor
escape_parser       memory3.c.save      process_pipeline    waitpid_monitor.c
escape_parser.c     memory4             process_pipeline.c
2520030537_SHELLFORGE$ pwd
/home/monis/OSLAB/skill
2520030537_SHELLFORGE$ date
Tue Sep 29 18:37:40 IST 2026
2520030537_SHELLFORGE$ cal
    September 2026
Su Mo Tu We Th Fr Sa
                1  2  3  4  5
 6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30

2520030537_SHELLFORGE$ ps
  PID TTY          TIME CMD
   310 pts/0        00:00:00 bash
   972 pts/0        00:00:00 myshell
   979 pts/0        00:00:00 ps
2520030537_SHELLFORGE$ exit
Exiting SHELLFORGE...
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

3. Demonstrate your Skill programs inside your shell

For example, suppose your Skill folder contains skill1.c. Inside your custom shell, type:

```
2520030537_SHELLFORGE$ gcc memory4.c -o memory4
2520030537_SHELLFORGE$ ./memory4
```

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill

```
error.txt          memory2.c          process_groups     variable_expansion
error_handling     memory3            process_groups.c   variable_expansion.c
error_handling.c   memory3.c          process_groups.c.save waitpid_monitor
escape_parser      memory3.c.save     process_pipeline   waitpid_monitor.c
escape_parser.c    memory4          process_pipeline.c

2520030537_SHELLFORGE$ pwd
/home/monis/OSLAB/skill
2520030537_SHELLFORGE$ date
Tue Sep 29 18:37:40 IST 2026
2520030537_SHELLFORGE$ cal
    September 2026
Su Mo Tu We Th Fr Sa
                1  2  3  4  5
 6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30

2520030537_SHELLFORGE$ ps
  PID TTY          TIME CMD
   310 pts/0    00:00:00 bash
   972 pts/0    00:00:00 myshell
   979 pts/0    00:00:00 ps
2520030537_SHELLFORGE$ exit
Exiting SHELLFORGE...
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./myshell
=====
Welcome to 2520030537 SHELLFORGE
=====
2520030537_SHELLFORGE$ gcc memory4.c -o memory4
2520030537_SHELLFORGE$ ./memory4
Linked List: 10 20 30
Memory released successfully.
2520030537_SHELLFORGE$
```